

Overview and Skeletal Role:

Vitamin D (calciferol) is a fat-soluble vitamin that promotes calcium absorption in the gut and maintains adequate serum calcium and phosphate concentrations to enable normal bone mineralization. It is also needed for bone growth and bone remodeling by osteoblasts and osteoclasts. Without sufficient vitamin D, bones can become thin, brittle, or misshapen.

Recommended Dietary Allowances (RDA) for Vitamin D:

- Infants 0-12 months: 400 IU (10 mcg) daily (Adequate Intake)
- Children and Adults 1-70 years: 600 IU (15 mcg) daily (includes pregnant and lactating females)
- Adults 71 years and older: 800 IU (20 mcg) daily

Tolerable Upper Intake Levels (UL) for Vitamin D:

The FNB established the following ULs:

- Infants 0-6 months: 1,000 IU (25 mcg) daily
- Infants 7-12 months: 1,500 IU (37.5 mcg) daily
- Children 1-3 years: 2,500 IU (62.5 mcg) daily
- Children 4-8 years: 3,000 IU (75 mcg) daily
- Children 9+ and Adults: 4,000 IU (100 mcg) daily

Sources of Vitamin D:

- Cutaneous Synthesis: The body synthesizes vitamin D₃ in the skin when exposed to ultraviolet B (UVB) radiation from sunlight. Season, latitude, air pollution, cloud cover, and melanin affect synthesis.
- Dietary Sources: Few foods naturally contain vitamin D. The best natural sources are flesh of fatty fish (such as trout, salmon, tuna, and mackerel) and fish liver oils. Beef liver, egg yolks, and cheese contain small amounts.
- Fortified Foods: In the U.S. food supply, fortified foods provide most dietary vitamin D, including fortified cow milk, fortified plant milk alternatives, infant formula, and fortified breakfast cereals.

Deficiency and Toxicity:

In children, severe vitamin D deficiency leads to rickets (failure of bone tissue to properly mineralize, resulting in soft bones and skeletal deformities). In adults, deficiency leads to osteomalacia and exacerbates osteoporosis. Excessive intake causes hypercalcemia, renal impairment, and vascular calcification.